Data-Efficient Time-Dependent PDE Surrogates: Graph Neural Simulators vs Neural Operators

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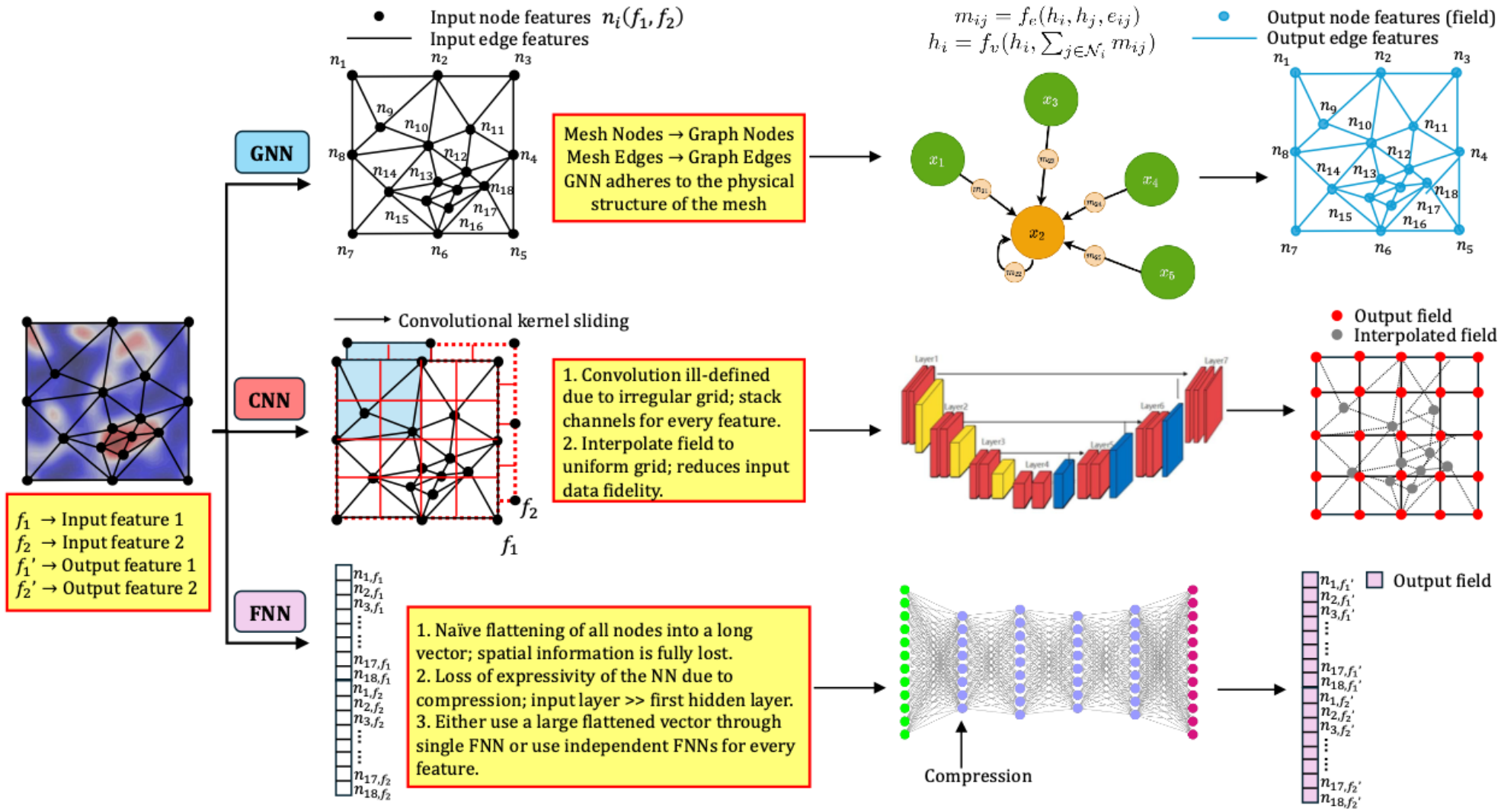
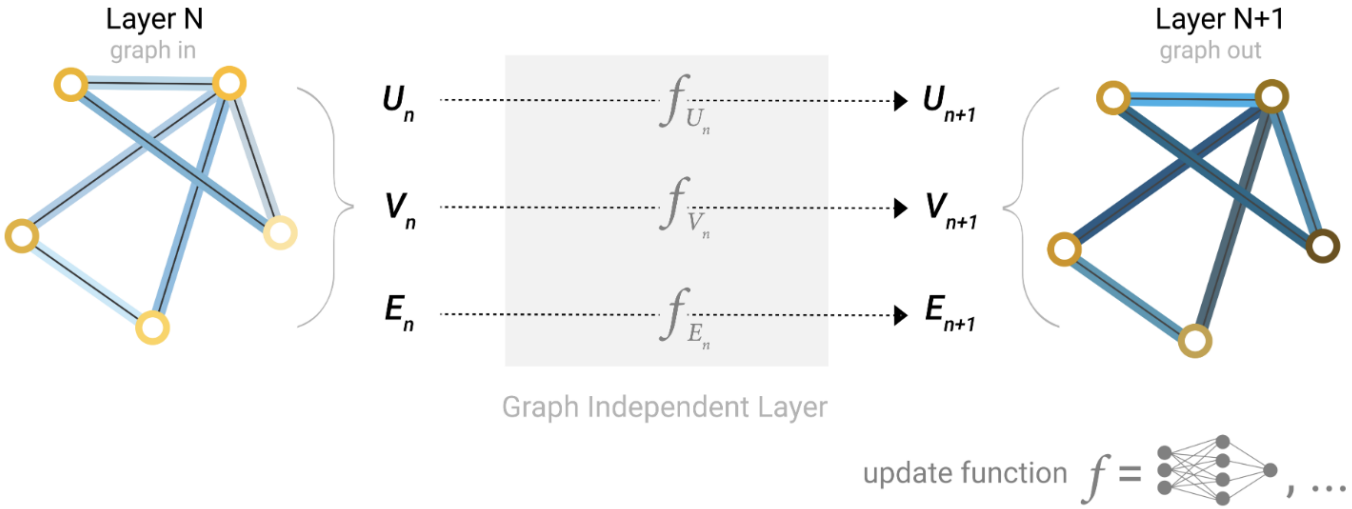
Learning in non-Euclidean spaces



Graph Neural Networks (GNNs)



- **Graph**: Abstract data structure represented as:

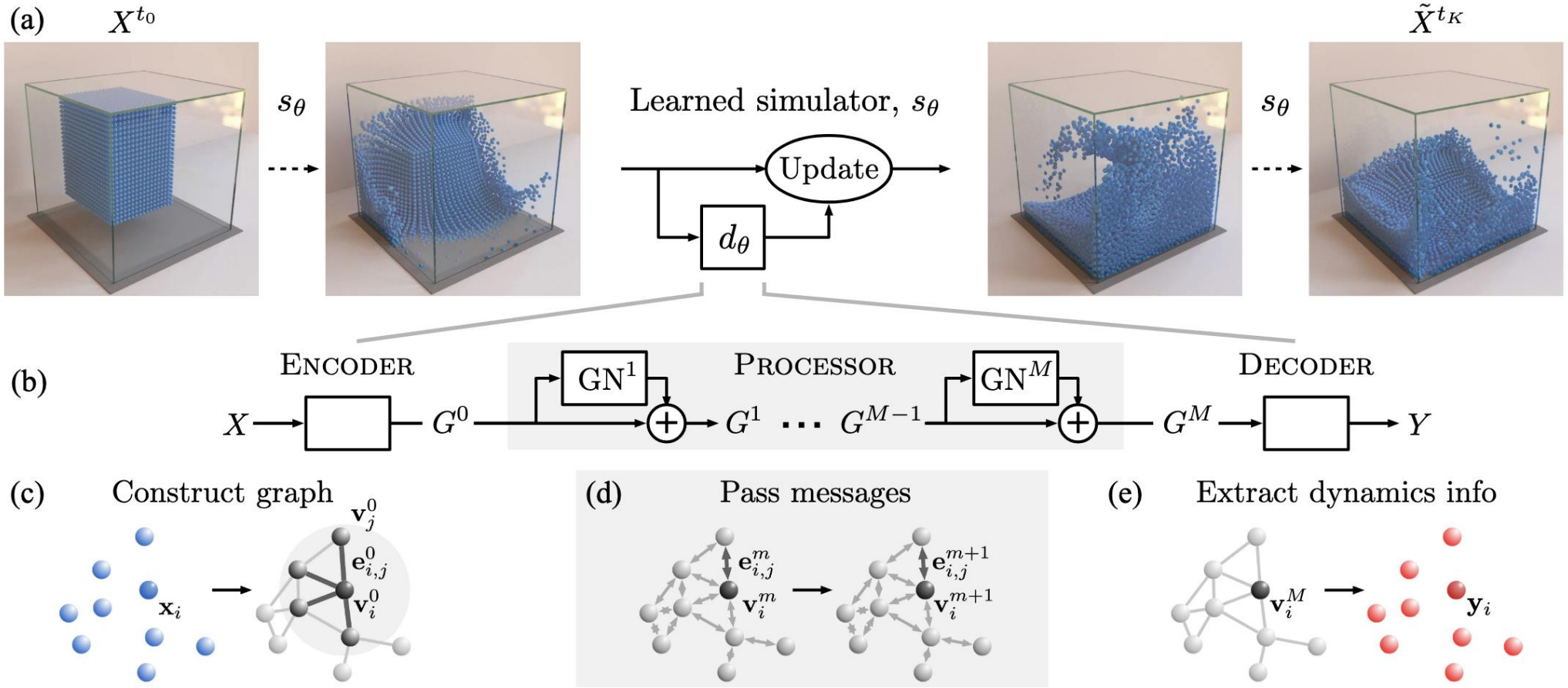
$$\mathcal{G} = (V, E, U),$$

where: (1) $V = \{\boldsymbol{v}_i\}$ are node features, (2) $E = \{(\boldsymbol{e}_{i,j}, i, j)\}$ are edge features with senders i and receivers j, (3) U is the global graph-level feature

- **GNNs:** Adopt "*Graph-in, Graph-out*" architecture meaning that the GNN takes a graph as an input and returns an output graph with an identical connectivity structure as the input graph.
  - Node, edge and global features are iteratively updated via neural networks (message passing):

$$V', E', U' = GNN(V, E, U)$$

Learn to simulate (L2S): Graph Network Simulators (GNS)



Sanchez-Gonzalez et al., *"Learning to simulate complex physics with graph networks"*, ICML (2020)

GNS - Mathematical Framework

- **Encoder**

$$v_i^t = MLP_v(x_i^t), \qquad e_{i,j}^t = MLP_e(r_{i,j}^t)$$

- **Processor (Message Passing)**

$$G = [G^1 \rightarrow G^2 \rightarrow \ldots \rightarrow G^M], \text{ where } G^{M+1} = GraphNetwork(G^M)$$

  - **Message Passing/Graph Network**

$$e'_{i,j} = m_{i,j} = MLP_{e'}\left(v_i, v_j, e_{i,j}\right) \text{ (Message Construction)}$$

$$\bar{v}_i = \sum_{j \in N(i)} e'_{i,j} \text{ (Aggregation of messages from neighbors – permutation invariant)}$$

$$v'_i = MLP\ (v_i, \bar{v}_i) \text{ (Node update)}$$

- **Decoder**

$$y_i^t = MLP_{v'}(v_i^{t'})$$

- **Updater**

$$\dot{p}_i^{t+1} = \dot{p}_i^t + y_i^t \cdot \Delta t$$
$$p_i^{t+1} = p_i^t + \dot{p}_i^{t+1} \cdot \Delta t$$

GNS as an autoregressive model for PDEs

- Let a solution field $u(x,t)$ be defined over a temporal domain $t \in [0,T]$ and a spatial domain $\boldsymbol{x} = [x_1, x_2, \dots, x_m] \in \mathcal{X} \subseteq \mathbb{R}^m$. The dynamics of u is governed by a PDE which can be represented as:

$$\frac{\partial \boldsymbol{u}}{\partial t} = \mathcal{F}(\boldsymbol{u}, \boldsymbol{u_x}, \boldsymbol{u_{xx}}, \dots, t)$$

- Along with a set of initial and boundary conditions defined as:

$$\boldsymbol{u}(\boldsymbol{x}, t = 0) = \boldsymbol{u_0}(\boldsymbol{x})$$
$$\boldsymbol{B}(\boldsymbol{u}) = f_{BC}(\boldsymbol{x}, t) \quad \forall \boldsymbol{x} \in \Gamma$$

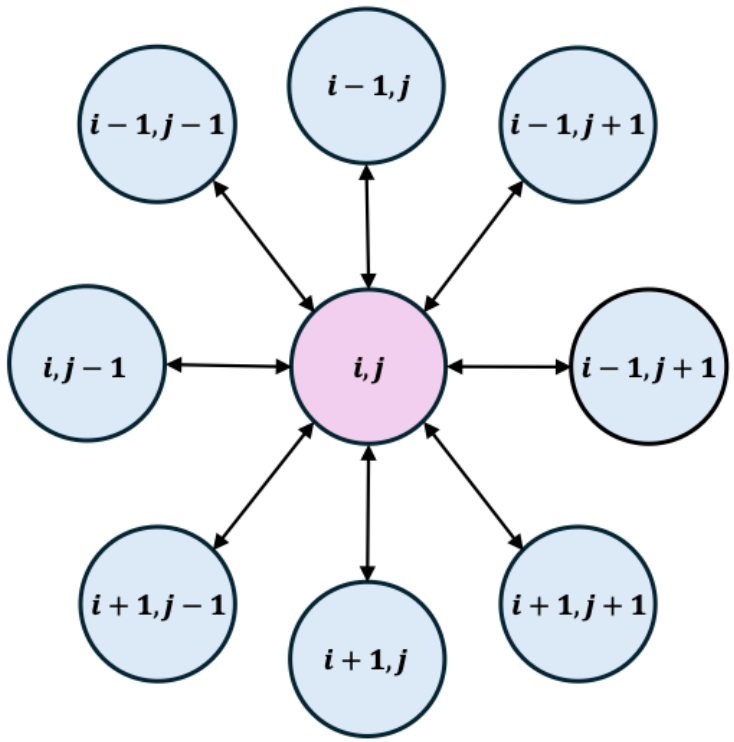
- The RHS time derivative $\mathcal{F}$ is majorly influenced by the spatial derivatives.
- Since, GNNs can leverage spatial interactions effectively, we can use them to learn $\mathcal{F}$ and then employ any numerical time integration scheme to evolve the solution in time in an autoregressive manner
- For example, an explicit Euler update results in:

$$\boldsymbol{u}(\boldsymbol{x}, t+1) = u(\boldsymbol{x}, t) + \Delta t \cdot GNN(\boldsymbol{u}(x,t), x, t)$$

Dibyajyoti Nayak and Somdatta Goswami, *"Data-Efficient Time-Dependent PDE Surrogates: Graph Neural Simulators vs Neural Operators"*, arXiv preprint arXiv:2509.06154 (2025)

Graph Construction

- **Nodes:**
  - Each grid point $x_{i,j} \in \mathbb{R}^2$, where $i = \{1, 2, \dots, N_x\}, j = \{1, 2, \dots, N_y\}$
  - Total number of nodes $(N) = N_x \times N_y$
- **Edges:**
  - Undirected edges to the k-nearest neighbors ($k = 8$ in our case)
  - Edge set: $\mathcal{E} = \{(u, v) \mid v \in \mathcal{N}_k(u)\}$
- **Nodes features:**
  - Physical fields of interest (e.g., velocity, pressure, material parameters)
  - Positional encoding: absolute coordinates, (x_u, y_u)
  - Fourier features: $[\sin(\omega x_u), \cos(\omega x_u), \sin(\omega y_u), \cos(\omega y_u)]$
- **Edge features:**
  - Relative displacement: $\Delta\mathbf{x}_{\mathrm{u,v}} = \mathbf{x}_{\mathrm{v}} - \mathbf{x}_{\mathrm{u}}$
  - Euclidean distance: $d_{\mathrm{u,v}} = \left\|\Delta\mathbf{x}_{\mathrm{u,v}}\right\|_2$
  - Field difference vector: $\Delta f_{\mathrm{u,v}} = f_{\mathrm{v}} - f_{\mathrm{u}}$
  - Norm of field difference vector: $\left\|\Delta f_{\mathrm{u,v}}\right\|_2$



2D Burgers' scalar field

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + u\frac{\partial u}{\partial y} = \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right),$$

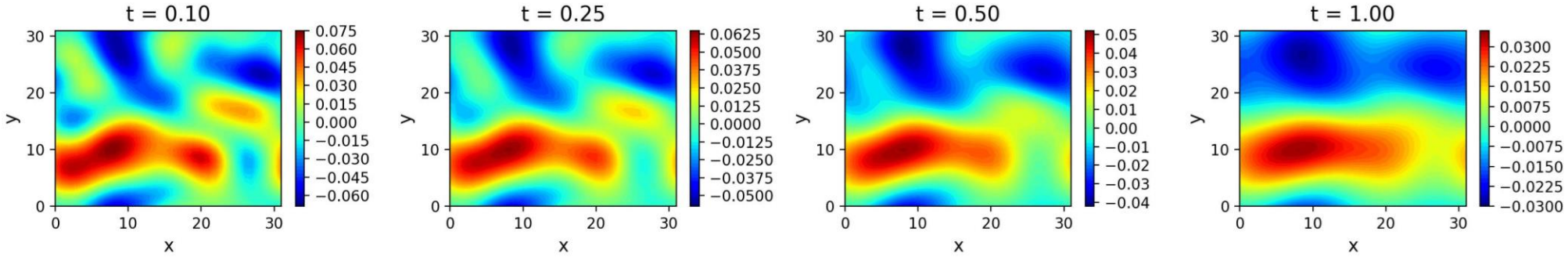
$$u(x, y, 0) = u_0(x, y)$$

$$x, y \in [0, 1), t \in [0, 1], \mu = 0.01$$

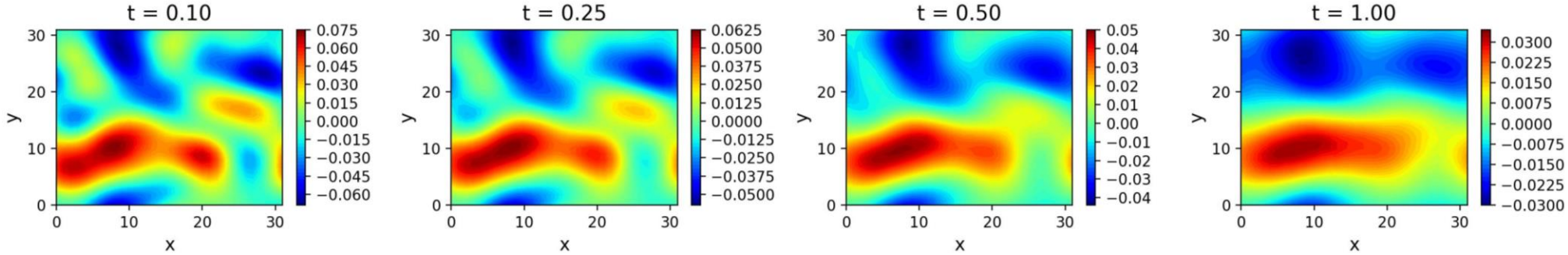
- **Initial conditions**: Gaussian random fields (GRF) with Matern kernel
- **Boundary conditions**: Periodic at both boundaries of the unit-square domain
- $\boldsymbol{N_s} = 1000$ samples, $\boldsymbol{N_t} = 101$ timesteps, $\boldsymbol{N_x} = \boldsymbol{N_y} = 32$ spatial grid points
- Relative L_2 error = $\frac{\left|\left|u_{pred} - u_{true}\right|\right|_2}{\left|\left|u_{true}\right|\right|_2}$
- **Characteristics of the PDE**:
  - Multi-dimensional nonlinear parabolic convective-diffusive system
  - Extension of the 1D viscous Burgers' Equation with scalar output field in two-dimensions

Results - u field

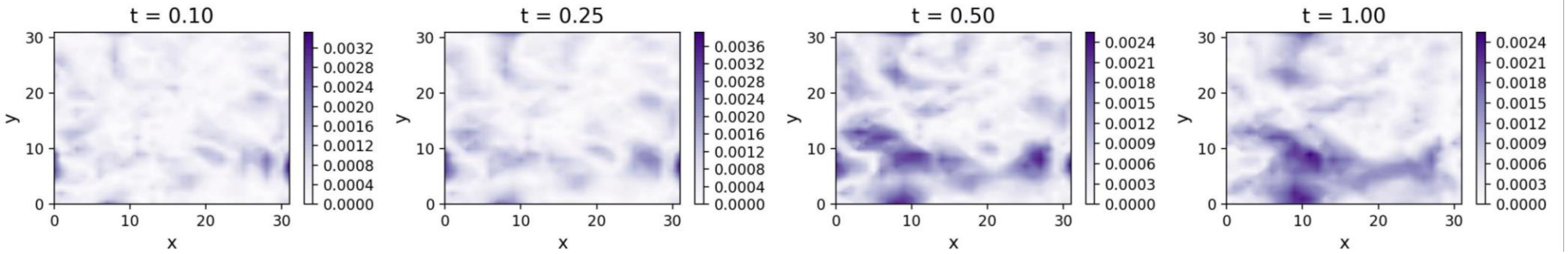
Ground Truth



Predicted



Absolute Error

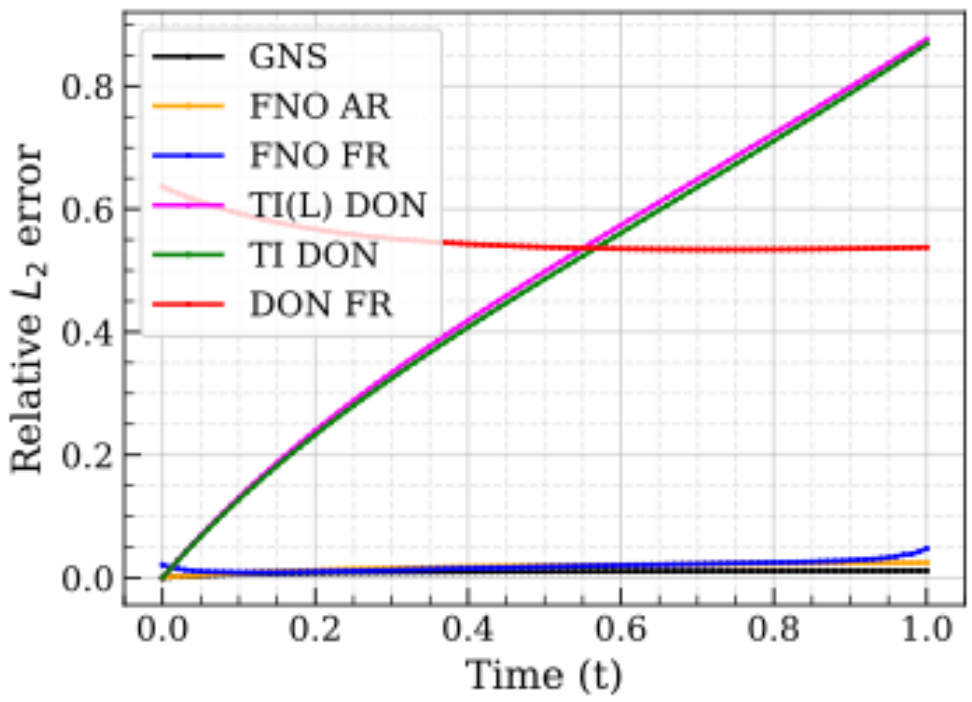


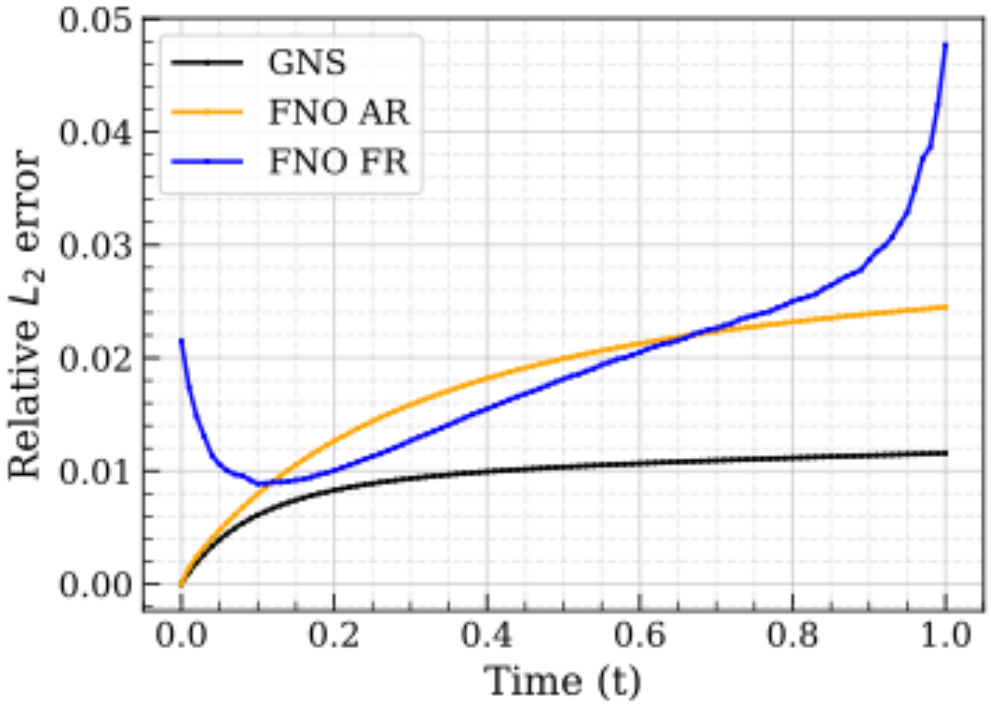
Generalization Study - Errors

Method	Training Trajectories			
	30	50	70	100
GNS	**0.0088**	**0.0045**	**0.00386**	**0.003626**
DON FR	0.56441	0.2892	0.20913	0.13424
DON AR	6.69263	6.55041	3.3983	1.16416
TI-DON	0.430894	0.29132	0.20473	0.17639
TI(L)-DON	0.43951	0.28119	0.2067	0.17559
FNO AR	0.01636	0.012934	0.006145	0.005266
FNO FR	0.017577	0.013386	0.010369	0.006383

- **GNS**: Shows superior performance across all methods, with relative L_2 errors below 1% even with 30 trajectories.
- **FNO**: Performs very well and closely follows the performance of GNS.
- **DeepONet-based frameworks**: Show higher generalization error, indicating that DeepONet is more susceptible to requiring larger amounts of data to effectively learn the underlying solution operator.

Error Accumulation





- **GNS**: Exhibits the lowest error growth with a noticeable plateauing of errors, indicating its ability to provide accurate, reliable, and stable performance over extended temporal horizons
- **FNO**: FNO AR largely performs better than FNO FR except from $t \in [0.1, 0.6]$
- **DeepONet-based frameworks**: Incur substantially higher errors than others due to being more data hungry. TI-based variants perform better than DON FR until $t = 0.5$, beyond which DON FR surpasses them

2D Coupled Burgers' Equation

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right)$$

$$x, y \in [0, 1), t \in [0, 1], \mu = 0.01$$

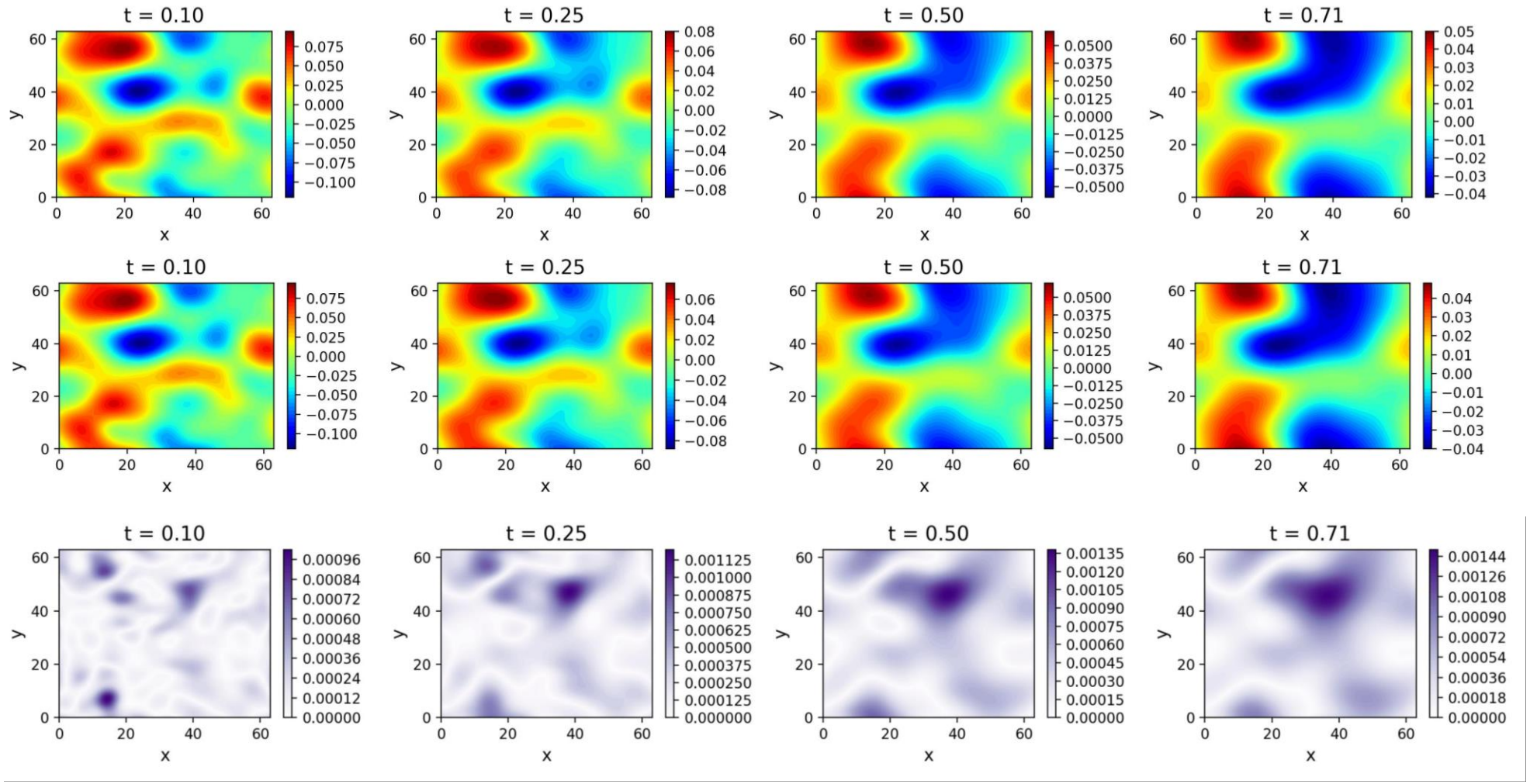
- **Initial conditions**: Gaussian random fields (GRF) with Matern kernel,
$$u(x, y, 0) = u_0(x, y), \qquad v(x, y, 0) = v_0(x, y)$$
- **Boundary conditions**: Periodic at both boundaries of the unit-square domain
- $\boldsymbol{N_s} = 500$ samples, $\boldsymbol{N_t} = 101$ timesteps, $\boldsymbol{N_x} = \boldsymbol{N_y} = 64$ spatial grid points
- Relative L_2 error = $\frac{\left|\left|u_{pred} - u_{true}\right|\right|_2}{\left|\left|u_{true}\right|\right|_2}$
- **Characteristics of the PDE**:
  - Represents nonlinear parabolic convection–diffusion of a vector field
  - Each component equation is coupled to the other via the cross-advection terms (vu_y, uv_x)

Results: u-velocity field

Ground Truth u

Predicted u

Error u

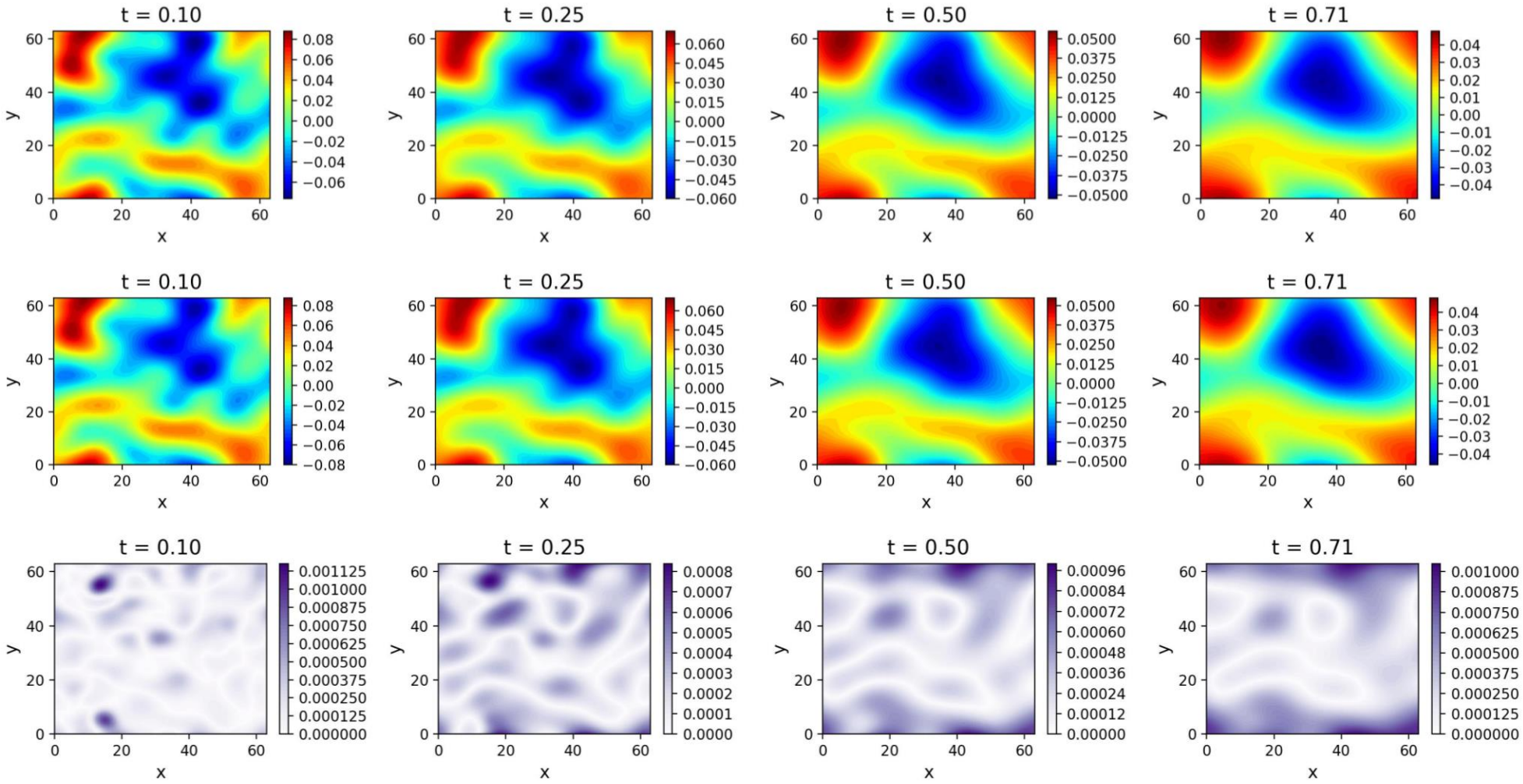


Results: v-velocity field

Ground Truth v

Predicted v

Error v

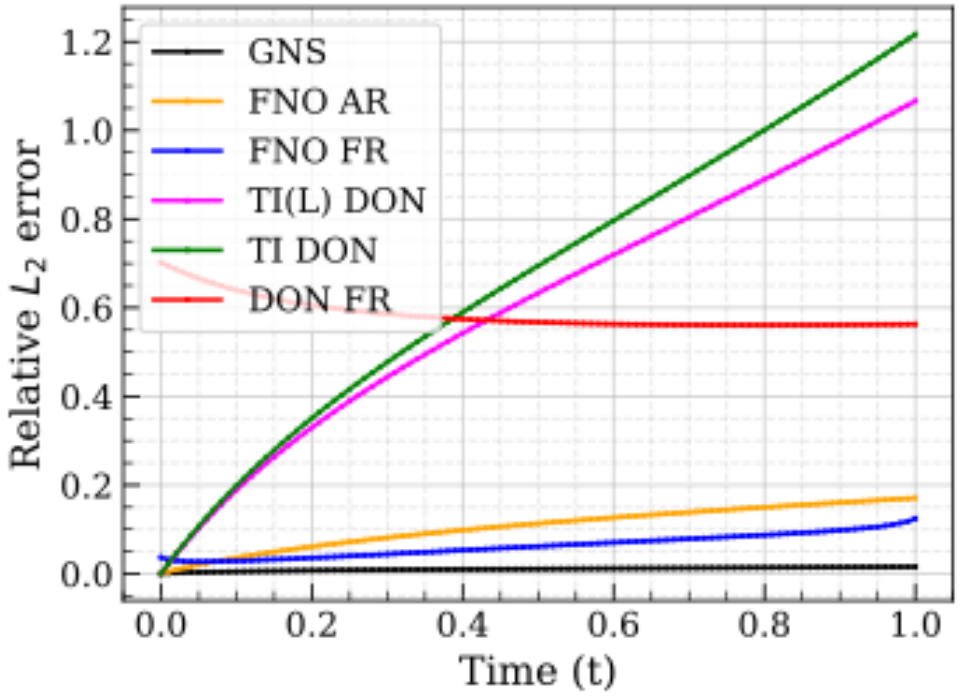


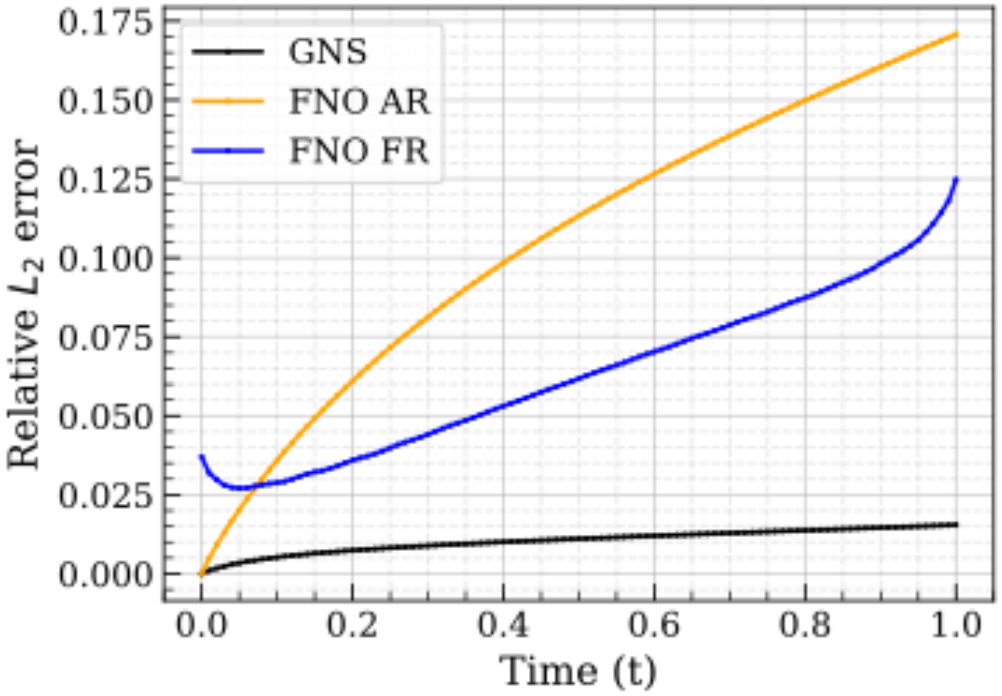
Generalization Study - Errors

Method	Training Trajectories			
	30	50	70	100
Field u				
GNS	**0.0091**	**0.00539**	**0.003431**	**0.00335**
DON FR	0.60742	0.4325	0.32581	0.2267
DON AR	3.2	8.259	1.0125	4.3491
TI-DON	0.535	0.3957	0.34607	0.2967
TI(L)-DON	0.52753	0.40503	0.34889	0.3081
FNO AR	0.087918	0.0408109	0.0227915	0.01426
FNO FR	0.054958	0.031109	0.023148	0.0149934
Field v				
GNS	**0.0078**	**0.00549**	**0.0036**	**0.003305**
DON FR	0.691209	0.3939	0.32882	0.2182
DON AR	5.593	3.713	12.6281	9.1233
TI-DON	0.553	0.3947	0.33249	0.2769
TI(L)-DON	0.56157	0.395902	0.33098	0.2764
FNO AR	0.098911	0.042962	0.024126	0.015502
FNO FR	0.0499403	0.0310453	0.0227212	0.0146658

- **GNS**: Shows superior performance across all methods, with relative L_2 errors below 1% even with 30 trajectories.
- **FNO**: Performs very well and closely follows the performance of GNS.
- **DeepONet-based frameworks**: Show higher generalization error, indicating that DeepONet is more susceptible to requiring larger amounts of data to effectively learn the underlying solution operator.

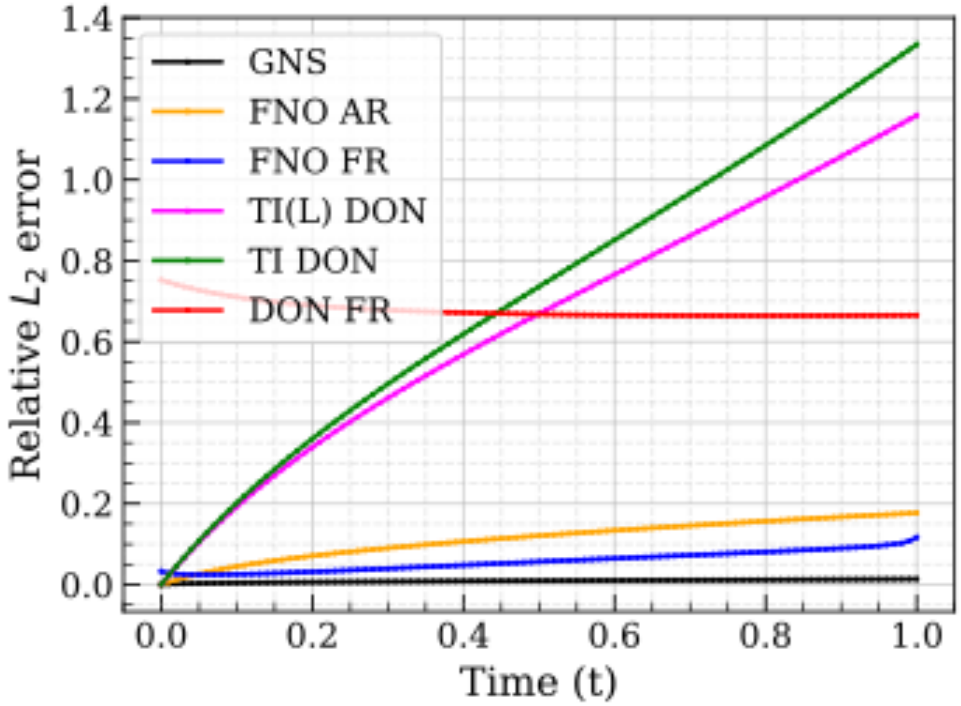
Error Accumulation - u-field

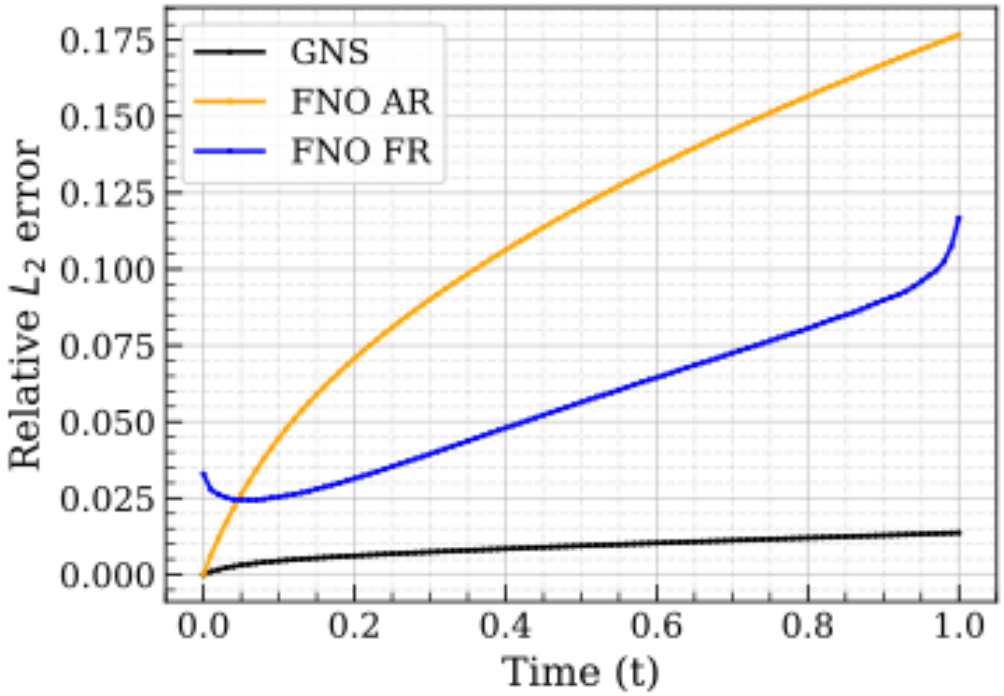




- **GNS**: Exhibits the lowest error growth with a noticeable plateauing of errors, indicating its ability to provide accurate, reliable, and stable performance over extended temporal horizons
- **FNO**: FNO FR performs better than FNO AR except for a few early timesteps until $t = 0.075$
- **DeepONet-based frameworks**: Incur substantially higher errors than others due to being more data hungry. TI-based variants perform better than DON FR until $t = 0.4$, beyond which DON FR surpasses them

Error Accumulation - v-field





- **GNS**: Exhibits the lowest error growth with a noticeable plateauing of errors, indicating its ability to provide accurate, reliable, and stable performance over extended temporal horizons
- **FNO**: FNO FR performs better than FNO AR except for a few early timesteps until $t = 0.05$
- **DeepONet-based frameworks**: Incur substantially higher errors than others due to being more data hungry. TI-based variants perform better than DON FR until $t = 0.5$, beyond which DON FR surpasses them

2D Allen-Cahn Equation

- The 2D Allen-Cahn PDE is defined as:

$$\frac{\partial u}{\partial t} = \epsilon^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) - (u^3 - u)$$

- Interfacial width or diffusion length: $\epsilon = 0.05$
- Initial conditions: GRF with Matern' Kernel

$$u(x, y, t = 0) = u_0(x, y)$$

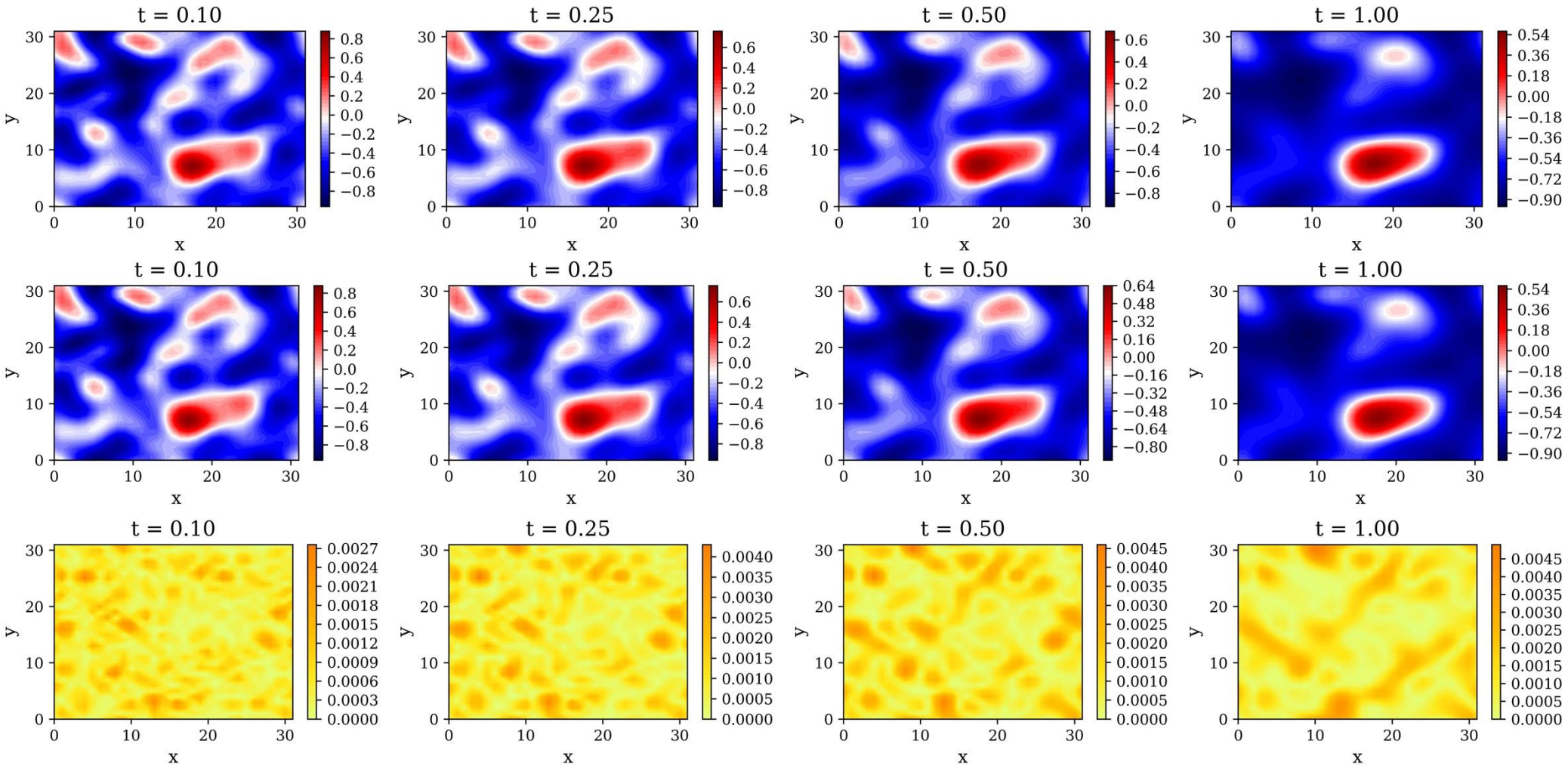
- **Boundary conditions**: Periodic at both boundaries of the unit-square domain
- $\boldsymbol{N_s} = 1000$ samples of ICs, $\boldsymbol{N_t} = 101$ timesteps, $\boldsymbol{N_x} = \boldsymbol{N_y} = 32$ spatial grid points
- **Characteristics of the PDE**:
  - Nonlinear parabolic PDE with bistable states
  - Interplay between a diffusion term ($\epsilon^2 \nabla^2 u$) and a reaction term ($-(u^3 - u)$)
  - Prototype model for phase separation and interface dynamics; L^2 gradient flow of the Ginzburg–Landau free energy functional

Results – u-field

Ground Truth

Predicted

Absolute Error

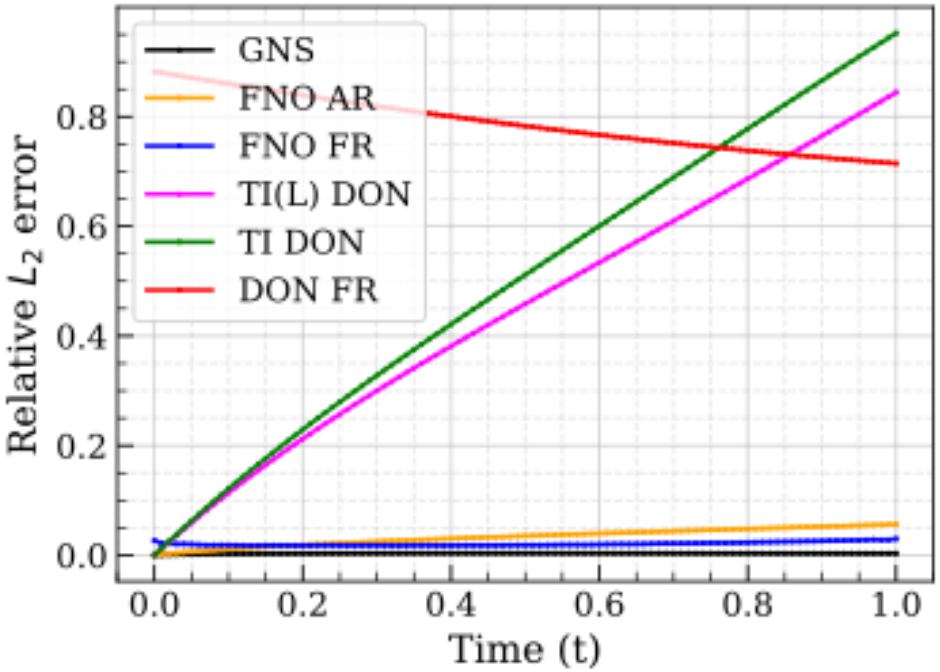


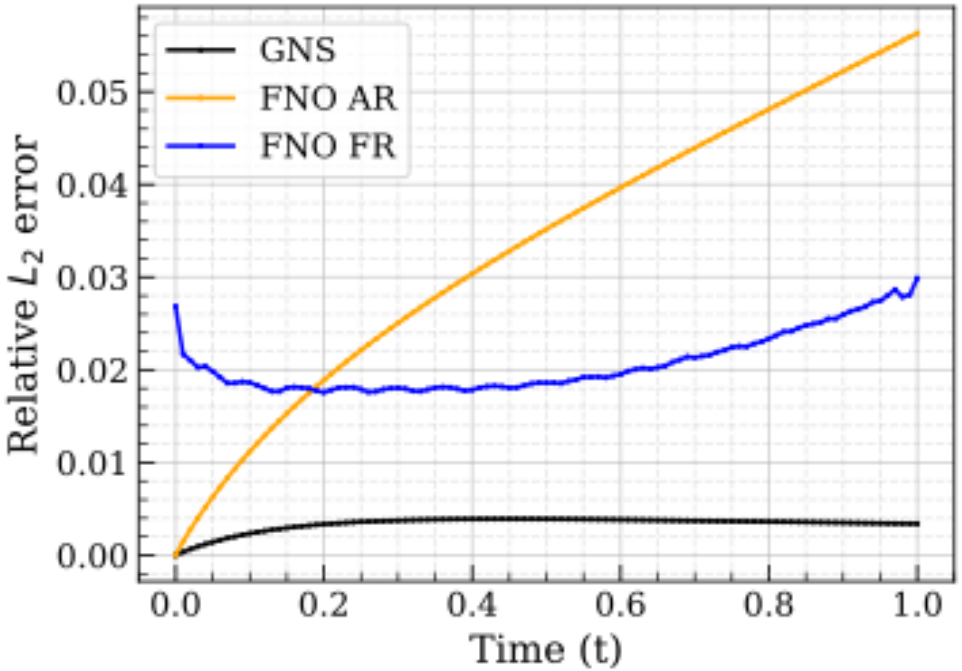
Generalization Study - Errors

Method	Training Trajectories			
	30	50	70	100
GNS	**0.003483**	**0.003437**	**0.003399**	**0.003394**
DON FR	0.78454	0.673335	0.632006	0.59262
DON AR	1.34333	1.24514	1.51555	1.65602
TI-DON	0.59872	0.596406	0.55184	0.52573
TI(L)-DON	0.531176	0.53339	0.51611	0.48485
FNO AR	0.038041	0.014089	0.008046	0.004988
FNO FR	0.0213076	0.0089356	0.0070546	0.0037004

- **GNS**: Shows superior performance across all methods, with relative L_2 errors below 0.4% even with 30 trajectories.
- **FNO**: Performs very well and closely follows the performance of GNS. Notably FNO-FR almost matches the inference accuracy of GNS with 100 training trajectories
- **DeepONet-based frameworks**: Show higher generalization error, indicating that DeepONet is more susceptible to requiring larger amounts of data to effectively learn the underlying solution operator.

Error Accumulation - u-field





- **GNS**: Exhibits the lowest error growth with a noticeable plateauing of errors, indicating its ability to provide accurate, reliable, and stable performance over extended temporal horizons
- **FNO**: FNO FR performs better than FNO AR except for a few early timesteps until $t = 0.15$
- **DeepONet-based frameworks**: Incur substantially higher errors than others due to being more data hungry. TI-based variants perform better than DON FR for most timesteps, except for the last few timesteps (i.e., $t \in [0.85, 1.0]$), beyond which DON FR surpasses them.

2D Nonlinear Shallow Water Equations

$$\frac{\partial \eta}{\partial t} + \frac{\partial (\eta u)}{\partial x} + \frac{\partial (\eta v)}{\partial y} = 0$$

$$\frac{\partial (\eta u)}{\partial t} + \frac{\partial}{\partial x}(\eta u^2 + \frac{1}{2} g \eta^2) + \frac{\partial (\eta u v)}{\partial y} = \mu(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2})$$

$$\frac{\partial (\eta v)}{\partial t} + \frac{\partial (\eta u v)}{\partial x} + \frac{\partial}{\partial y}(\eta v^2 + \frac{1}{2} g \eta^2) = \mu(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2})$$

$$\eta(x, y, 0) = \eta_0(x, y), \qquad u(x, y, 0) = u_0(x, y), \qquad v(x, y, 0) = v_0(x, y)$$

$$x, y \in [0, 1), t \in [0, 1], \mu = 0.002, g = 1$$

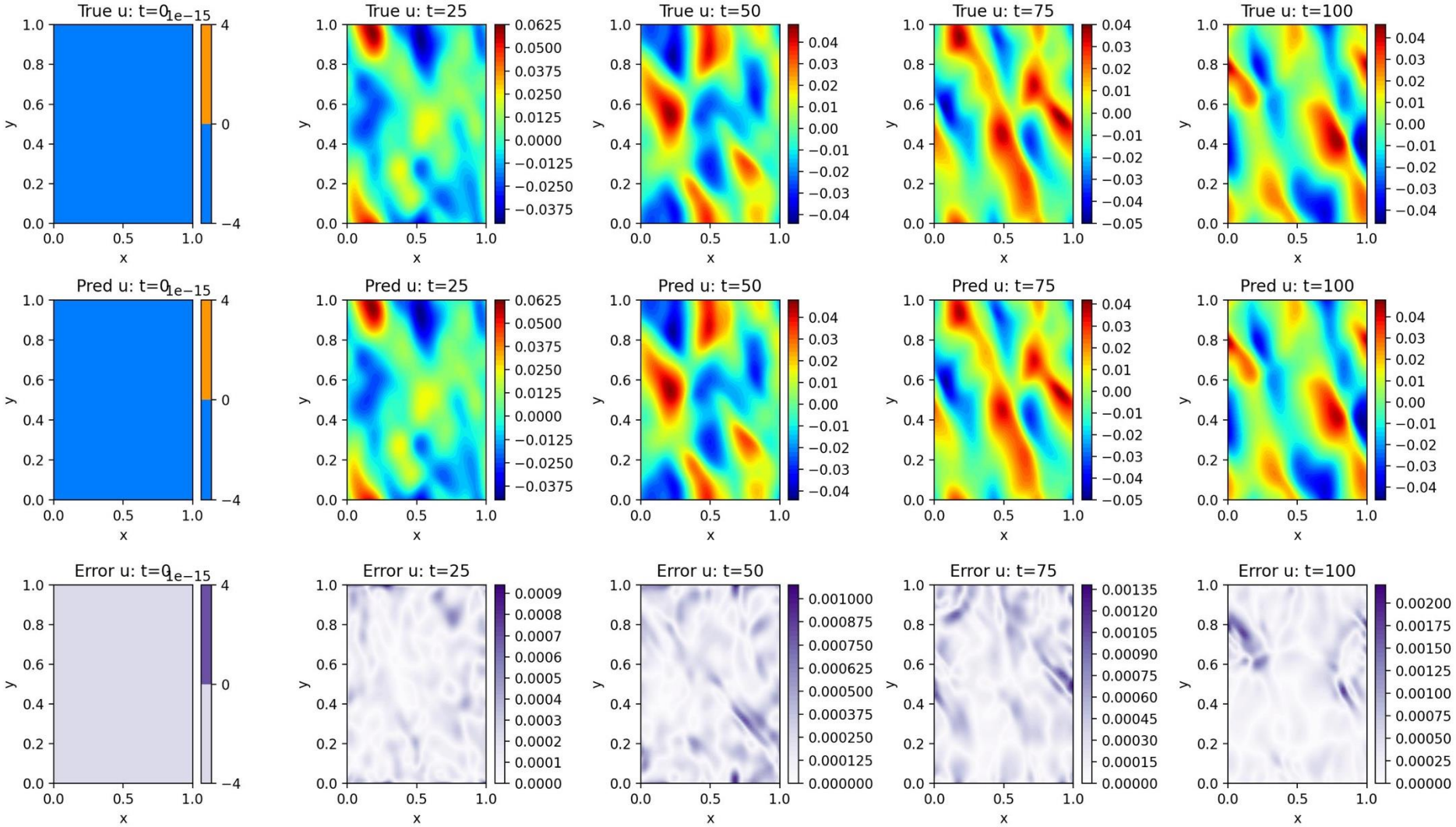
- $\eta(x, y, t)$: Height of the perturbed surface, $u(x, y, t), v(x, y, t)$: Velocity fields in the x and y directions
- **Boundary conditions**: Periodic at both boundaries of the unit-square domain
- $\boldsymbol{N_s} = 500$ samples, $\boldsymbol{N_t} = 101$ timesteps, $\boldsymbol{N_x} = \boldsymbol{N_y} = 64$ spatial grid points
- **Characteristics of the PDE**:
  - Nonlinear hyperbolic system of conservation laws exhibiting wave-like solutions with sharp gradients, where nonlinear advection, pressure gradients, and source terms drive gravity waves, shocks, and hydraulic jumps.

Results: u-velocity field

Ground Truth u

Predicted u

Error u

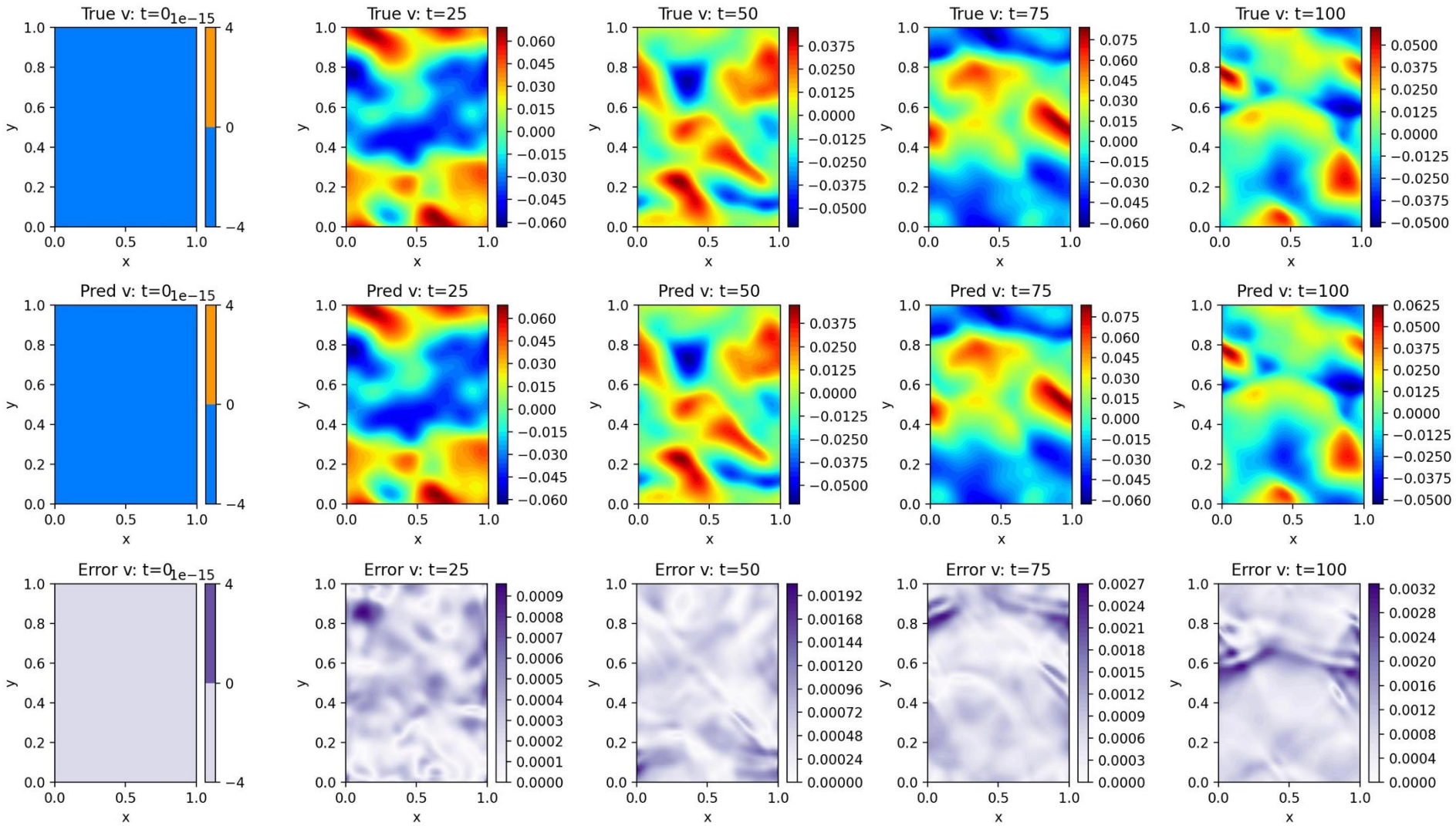


Results: v-velocity field

Ground Truth v

Predicted v

Error v

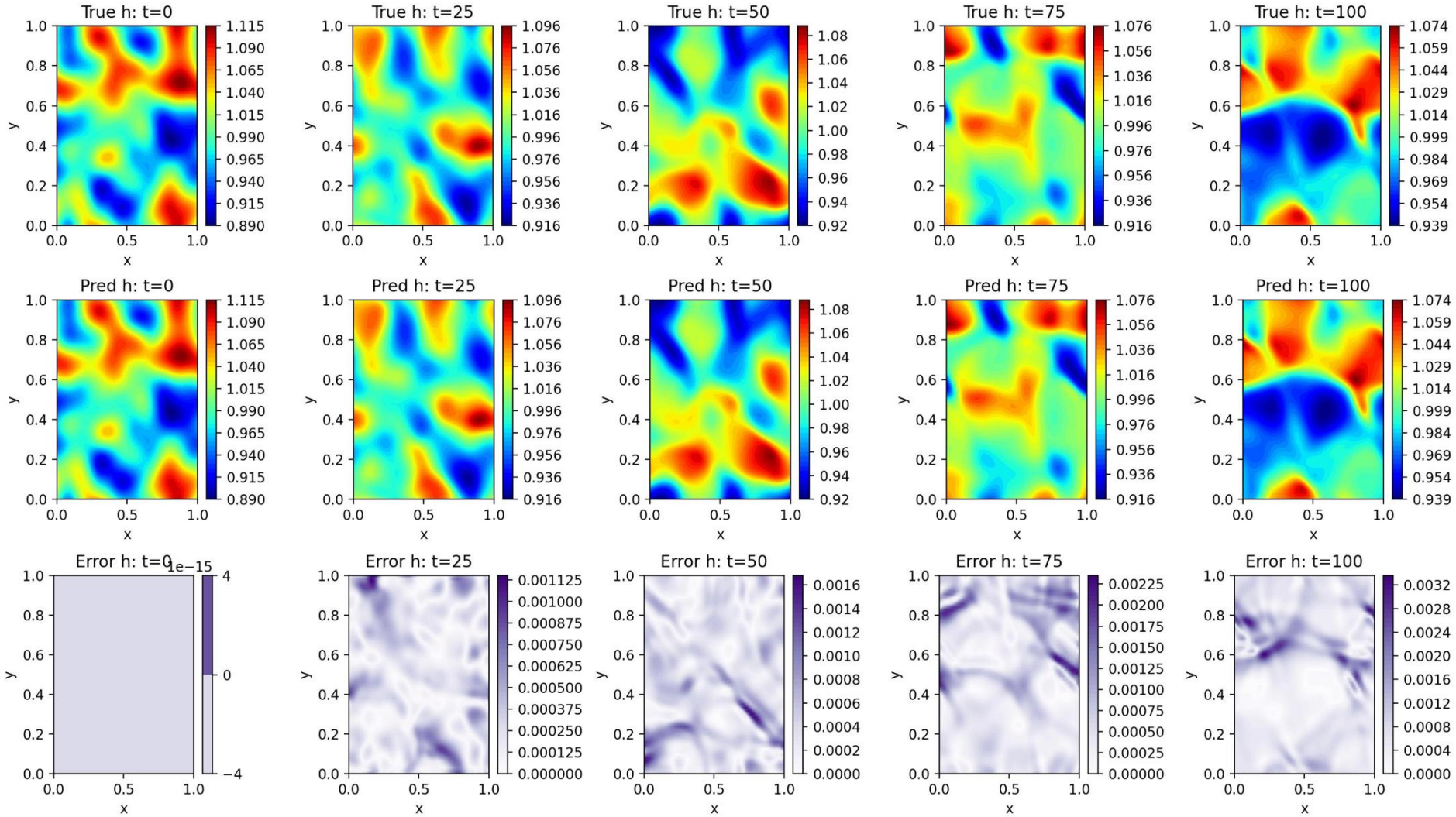


Results: height field (h)

Ground Truth h

Predicted h

Error h

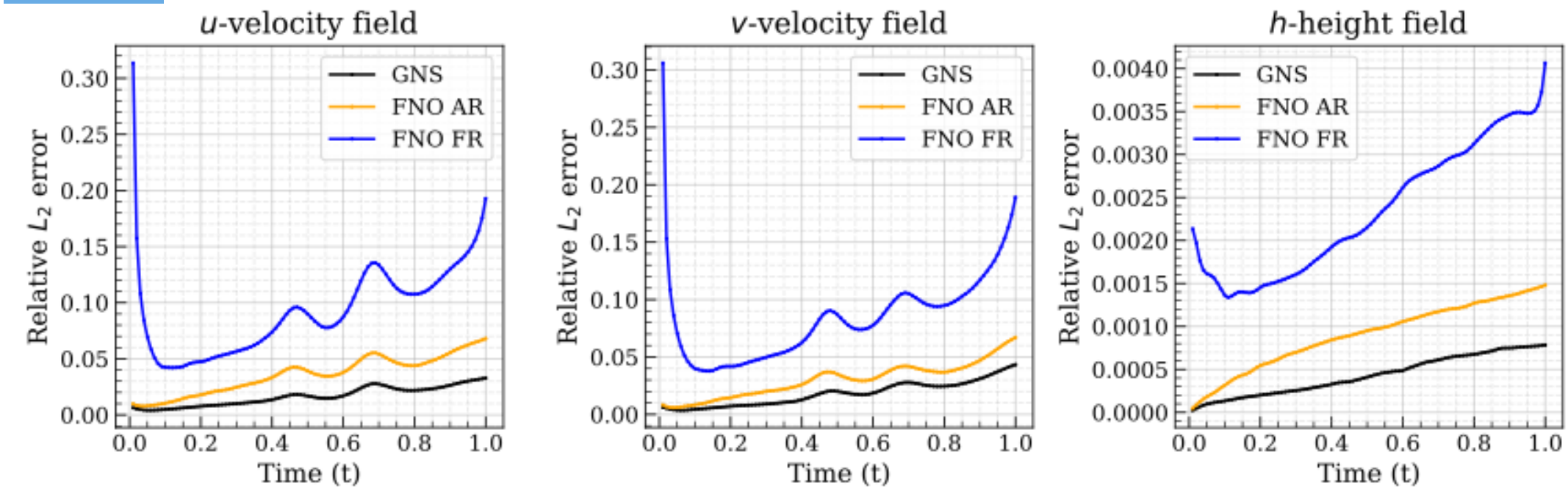


Generalization Study - Errors

Method	Training Trajectories			
	30	50	70	100
Field u				
GNS	**0.0323**	**0.01718**	**0.01175**	**0.00505**
FNO AR	0.0347	0.03682	0.02904	0.02366
FNO FR	0.1496	0.09056	0.06069	0.04432
Field v				
GNS	**0.0270**	**0.01916**	**0.00933**	**0.00468**
FNO AR	0.03562	0.03098	0.02373	0.01947
FNO FR	0.1343	0.08105	0.05415	0.03989
Field h				
GNS	**0.00074**	**0.00048**	**0.00021**	**0.00012**
FNO AR	0.00102	0.00097	0.00076	0.00062
FNO FR	0.00411	0.00246	0.00162	0.00117

- **GNS**: Shows superior performance across all methods, with relative L_2 errors below 0.4% even with 30 trajectories.
- **FNO**: Performs very well and closely follows the performance of GNS. Notably FNO-AR almost matches the inference accuracy of GNS for 30 training trajectories
- **DeepONet-based frameworks**: In this case, the DeepONet-based frameworks incurred higher errors than usual (>50%) due to near-zero initial conditions (exact zero for both u and v velocity fields)

Error Accumulation - u, v, h-fields



- **GNS:** Exhibits the lowest error growth with a noticeable plateauing of errors, indicating its ability to provide accurate, reliable, and stable performance over extended temporal horizons.
- **FNO AR:** Performs very well, closely following the performance of GNS, but has relatively higher error than GNS over extended temporal horizons.
- **FNO FR:** Incurs high errors during the initial few timesteps for the u and v velocity fields (dynamics near zero). After t = 0.1, the errors decrease to a reasonably low level, but they once again grow over time.

Conclusions

- GNS can accurately model the evolution dynamics of PDEs with coupled nonlinear fields, even with limited training data.
- This is due to its inherent data representation as a graph. For example, in the 2D SWE case:
  - Input node feature vectors: $[u(x, y, t^n), v(x, y, t^n), h(x, y, t^n), static\ features]$
  - Output node feature vector: $[\dot{u}(x, y, t^n), \dot{v}(x, y, t^n), \dot{h}(x, y, t^n)]$
- This strong physics-aware inductive bias makes GNS an accurate and reliable PDE surrogate.
- Among the neural operators, DeepONet is more susceptible to error because it is more data-hungry.
- FNO alleviates this problem by leveraging spectral-space learning, but it may still require more data if the underlying PDE solution has a wide frequency spectrum.

Thank You!
Questions?